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Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 6 – Réseaux I

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

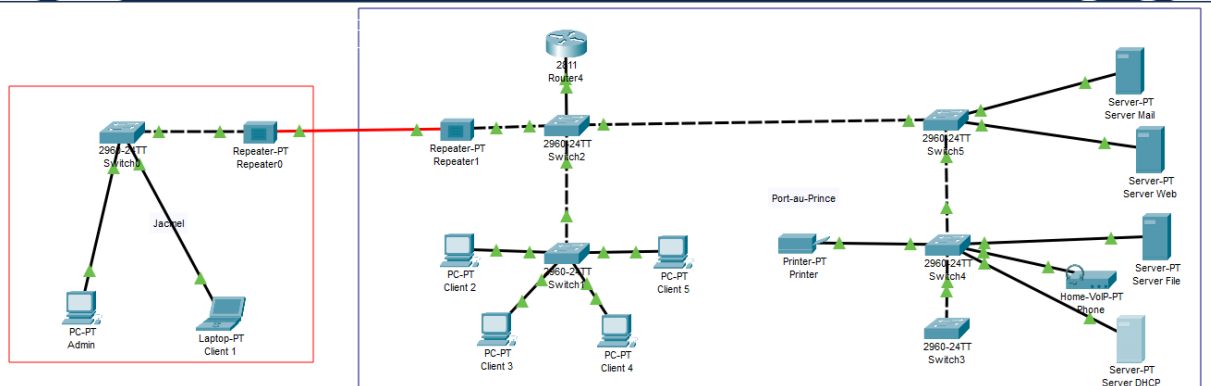
Niveau : L3

Date: 06/12/2025

L'objectif de ce TD est de :

1. Comprendre le fonctionnement du DHCP dans un réseau multi-LAN.
2. Mettre en place un serveur DHCP centralisé.
3. Configurer un routeur pour relayer les requêtes DHCP entre réseaux différents.
4. Configurer le service DHCP directement sur un routeur.
5. Vérifier l'attribution automatique d'adresses IP dans chaque LAN.
6. Comprendre l'adressage IPv6 et le rôle de DHCPv6.
7. Configurer un routeur comme serveur DHCPv6.
8. Attribuer automatiquement des adresses IPv6 aux hôtes.
9. Découvrir l'attribution automatique d'adresses IPv6 via DHCPv6.
10. Configurer un serveur DHCPv6 dans un seul réseau local.
11. Vérifier la communication entre les réseaux en IPv6

- 1) Reproduisons cette topologie en configurant les services DHCP afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau :



➤ Configuration du serveur DHCP

Server DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

Start IP Address: 192 168 1 3

Subnet Mask: 255 255 255 0

Maximum Number of Users: 30

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.1.1	0.0.0.0	192.168.1.3	255.255.255.0	30	0.0.0.0	0.0.0.0

☐ Top

➤ Ajout automatique des ip address ipv4 provenant serveur DHCP

Admin

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:64FF:FEA0:28AA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Client 1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.5

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FEC8:5753

Default Gateway

DNS Server

802.1X

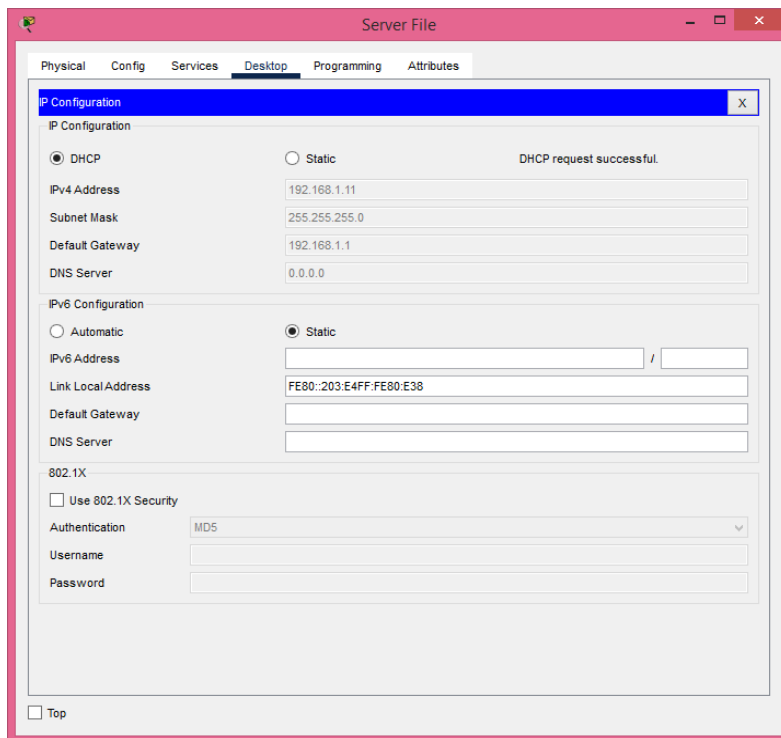
☐ Use 802.1X Security

Authentication MD5

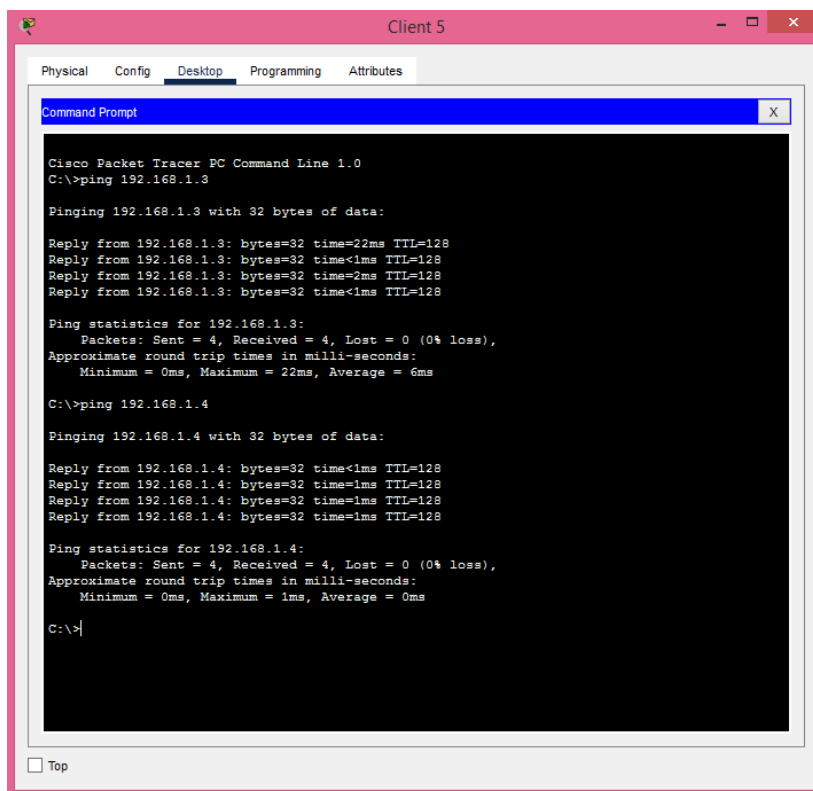
Username

Password

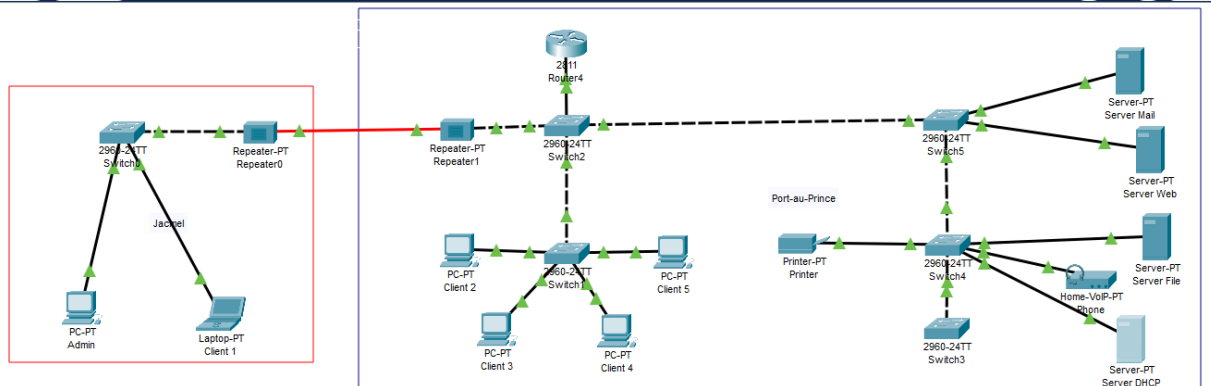
☐ Top



➤ Testons la connectivité avec ping



2. Reproduisons cette topologie en configurant les services DHCPv6 afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.



➤ Configuration du serveur DHCPv6

Server DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6**
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCPv6

Interface: FastEthernet0 Service: ☒ On ☐ Off

DHCPv6 Pool: 1

DHCPv6 Pool:

Pool List: 1 Create Pool Remove Pool

DNS Server: Domain Name:

IPv6 Address Prefix

Prefix	Prefix Length	Valid Lifetime	Preferred Lifetime
< [Progress Bar] >			

Create Edit Remove

IPv6 Prefix-Delegation

Prefix	DUID	Local Pool	Valid Lifetime	Preferred Lifetime

Create Edit Remove

IPv6 Local Pool

Pool Name	Prefix	Prefix Length

Create Edit Remove

☐ Top

➤ Ajout automatique des ip address ipv6 par le serveur DHCPv6

Admin

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.4

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:F1E9:9D6D:C8C1:C8D1 / 64

Link Local Address: FE80::201:64FF:FEA0:28AA

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Client 1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.5

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:DE9E:9E3:A566:3457 / 64

Link Local Address: FE80::2D0:BAFF:FEC8:5753

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

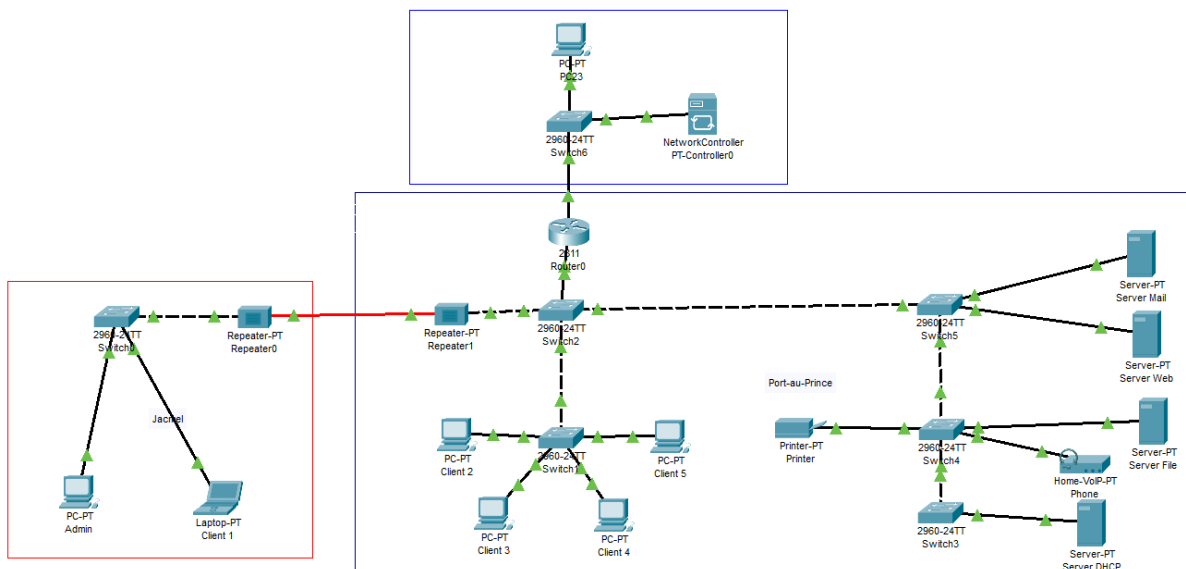
Authentication: MD5

Username:

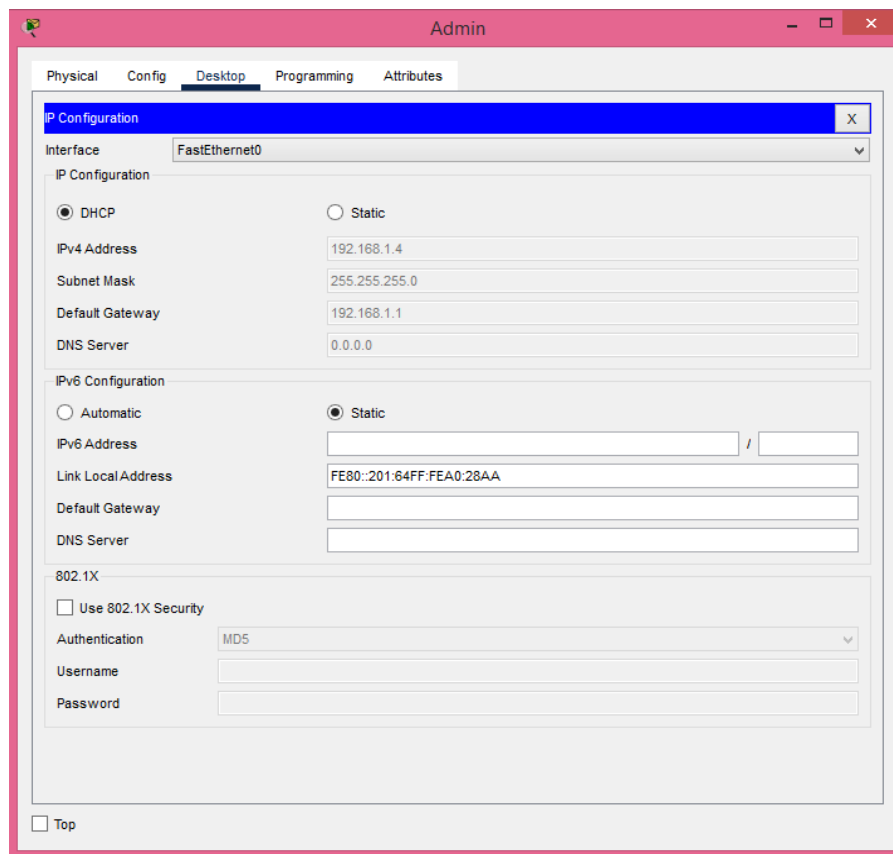
Password:

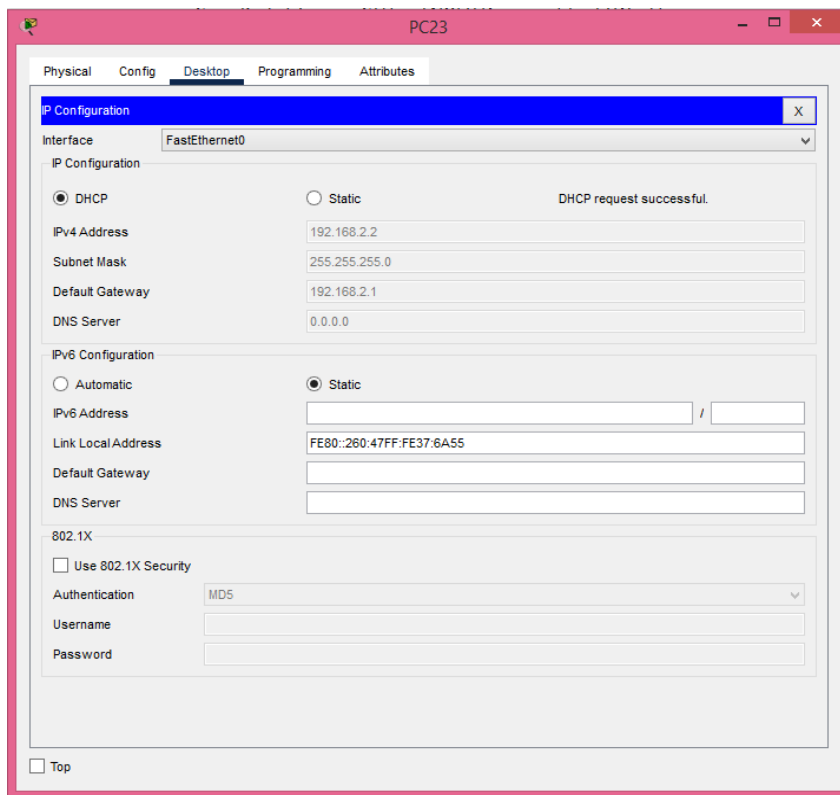
☐ Top

3. Reproduisons cette topologie en configurant les services DHCP, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.

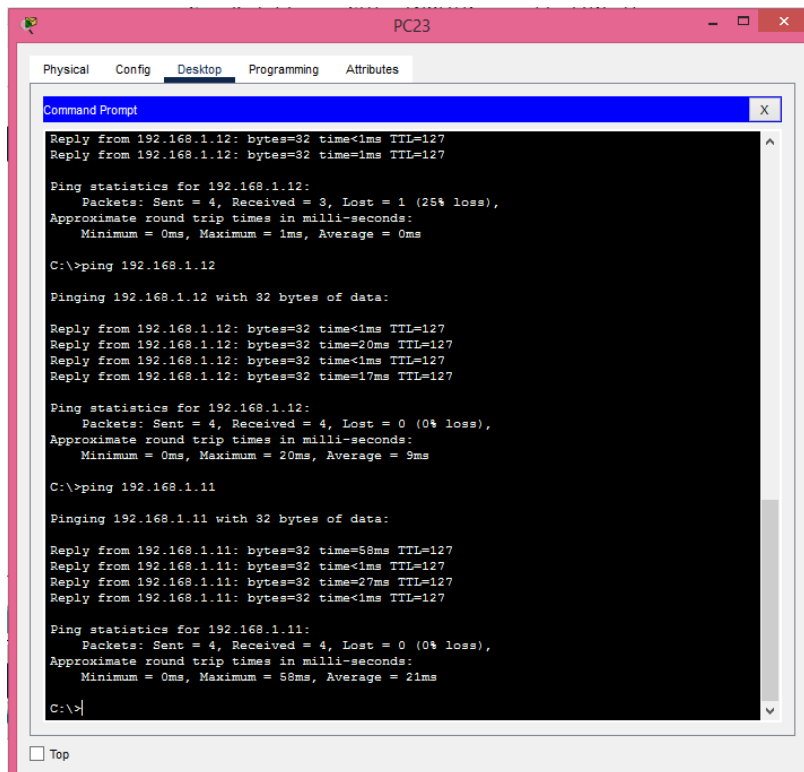


➤ **Ajout automatique des ip address ipv4 par le router utilisé comme serveur DHCP.**

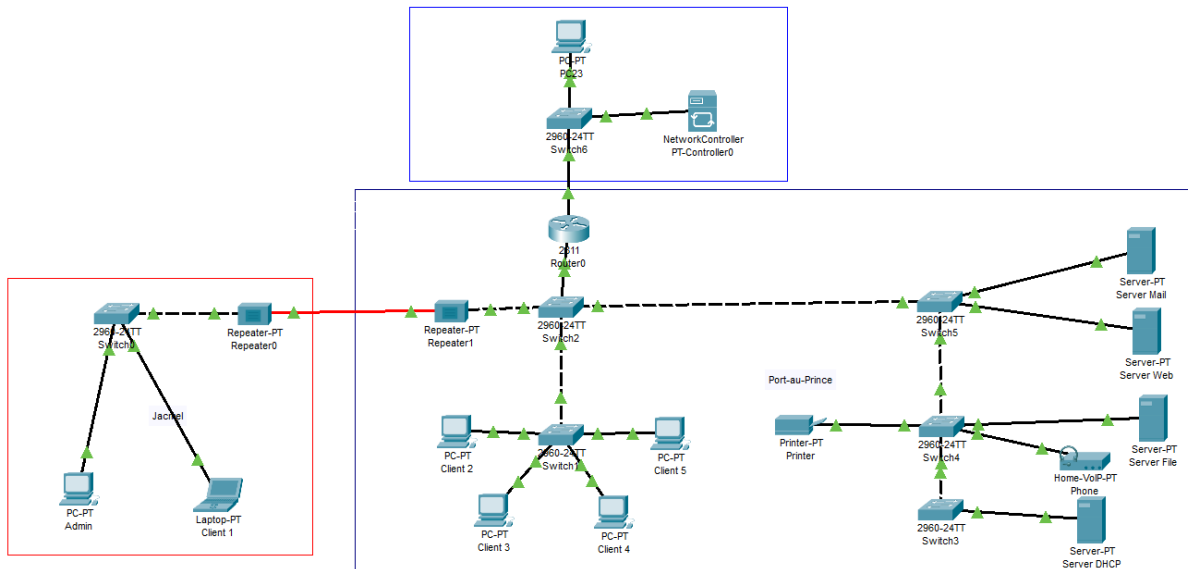




➤ *Testons la connectivité avec ping*



4. Reproduisons cette topologie en configurant les services DHCPv6, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.



- Ajout automatique des ip address ipv6 par le router utilisé comme serveur DHCPv6.

Client 3

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:2A7F:55D4:F247:9ECA / 64

Link Local Address: FE80::2E0:F9FF:FECD:5690

Default Gateway: FE80::2D0:FFFF:FE15:1D01

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Client 2

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.1.9

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address: 2001:DB8:1:0:1A7A:A85B:D4B0:D4B0 / 64

Link Local Address: FE80::290:2BFF:FE44:B2ED

Default Gateway: FE80::2D0:FFFF:FE15:1D01

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Server Mail

Physical Config Services Desktop Programming Attributes

IP Configuration X

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address	192.168.1.13
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static Ipv6 request successful.

IPv6 Address	2001:DB8:1:0:82C2:BD26:599A:E88B / 64
Link Local Address	FE80::20C:CFFF:FE51:BB5D
Default Gateway	FE80::2D0:FFFF:FE15:1D01
DNS Server	

802.1X

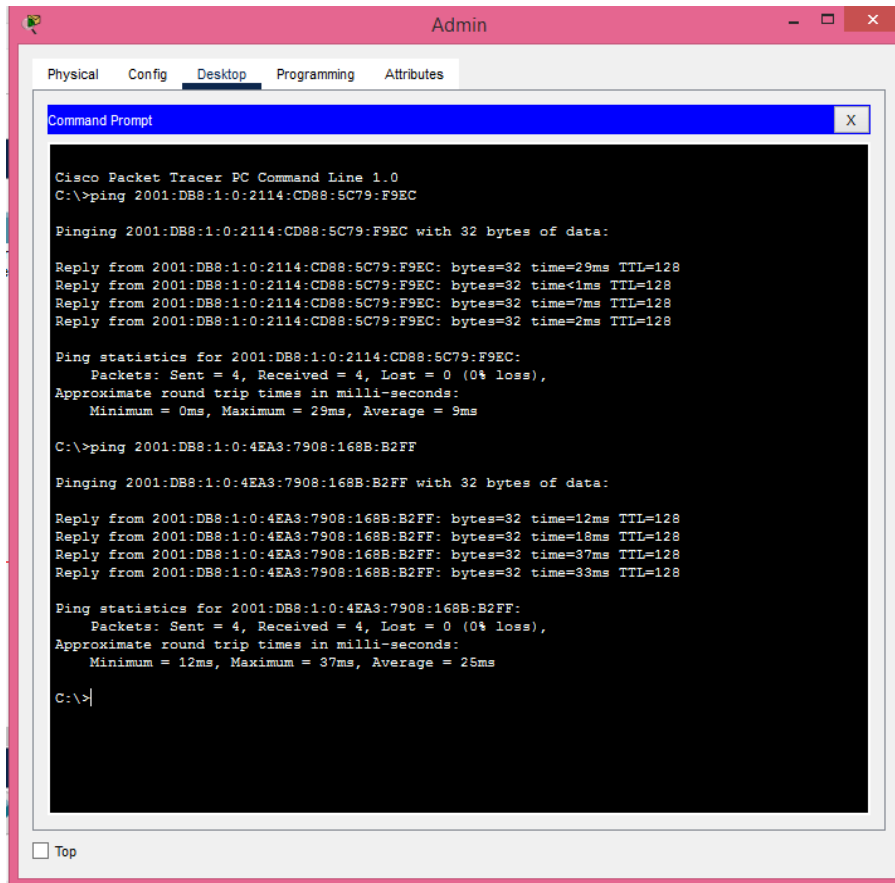
☐ Use 802.1X Security

Authentication: MD5 ▼

Username:

Password:

➤ Testons la connectivité avec ping



The screenshot shows a Cisco Packet Tracer PC Command Line window titled "Admin". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. Inside the window is a "Command Prompt" window with a blue title bar and a close button. The command prompt displays the output of two ping commands. The first command is "C:\>ping 2001:DB8:1:0:2114:CD88:5C79:F9EC", which results in four successful replies with 32 bytes of data, times ranging from 1ms to 29ms, and a TTL of 128. The statistics show 4 packets sent, 4 received, and 0 lost (0% loss), with an average round trip time of 9ms. The second command is "C:\>ping 2001:DB8:1:0:4EA3:7908:168B:B2FF", which also results in four successful replies with 32 bytes of data, times ranging from 12ms to 37ms, and a TTL of 128. The statistics show 4 packets sent, 4 received, and 0 lost (0% loss), with an average round trip time of 25ms. The command prompt ends with "C:\>|".

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1:0:2114:CD88:5C79:F9EC

Pinging 2001:DB8:1:0:2114:CD88:5C79:F9EC with 32 bytes of data:

Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=29ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=7ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=2ms TTL=128

Ping statistics for 2001:DB8:1:0:2114:CD88:5C79:F9EC:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 29ms, Average = 9ms

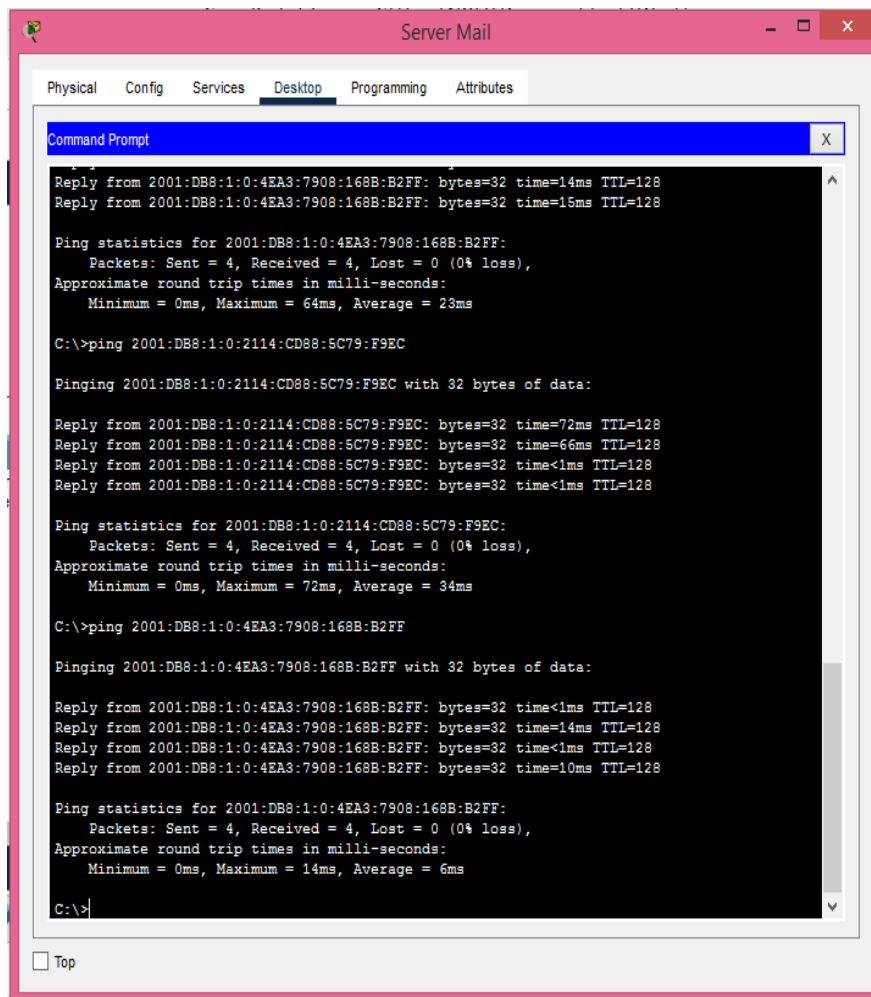
C:\>ping 2001:DB8:1:0:4EA3:7908:168B:B2FF

Pinging 2001:DB8:1:0:4EA3:7908:168B:B2FF with 32 bytes of data:

Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=12ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=18ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=37ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=33ms TTL=128

Ping statistics for 2001:DB8:1:0:4EA3:7908:168B:B2FF:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 37ms, Average = 25ms

C:\>|
```



Conclusion

Ce TD m'a permis de comprendre en profondeur le fonctionnement du protocole DHCP dans un environnement multi-LAN, en mettant en place un serveur DHCP centralisé ainsi qu'un service DHCP directement sur un routeur.

Ce TD m'a aussi permis d'aborder l'adressage automatique des hôtes via DHCPv6, de configurer un routeur comme serveur DHCPv6, et de gérer l'attribution d'adresses IPv6 dans un réseau local.